

#	Risk(s) identified	What might be affected and how	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)
1	Poor light diffusion	Inadequate engraving or surface finish may result in uneven illumination, reducing visibility and display quality.	Test samples,adjust engraving depth/patern	4	4	16
2	Arduino code errors	Bugs may result in incorrect LED sequencing or failure to display intended images.	Test in modules, thorough debugging	4	4	16
3	Soldering defects	Cold joints or bridges ,any cause intermittent or failed LED operation, reducing reliability of the display.	Inpection, reflow, continuity testing	3	4	12
4	Incorrect LED orientation	Reversed polarity prevents LEDs from lighting causing partial / total display failure	Compare to PCB schematic	4	3	12
5	Fire due to incorrect LED resistor values	Wrong current-limiting resistors may cause dim output or LED burnout, damaging components and risking a fire	Circuit verification, calculation checks	3	4	12
6	Acrylic sheets incompatible with MDF frame	Incorrect dimensions may prevent proper fitting into MDF structure, delaying assembly	Follow constraints, measure accurately	4	3	12
7	Acrylic damage	Aura face damage may scatter light incorrectly or reduce visual quality of the display	Keep in film until necessary, careful handling	4	3	12
8	Burning skin on hot LEDs	Contact with the LEDs (~50 C)during / after operation may cause skin burns.	Do not touch components, during/after operation, awareness	4	3	12
9	Contact with solder residue (ingestion risk)	Handling solder can leave residues on hands, which can be ingested if hands are not washed	Wash hands after handling, no eating/drinking in lab, use lead-free solder if possible	4	3	12
10	Cuts from sharp acrylic edge	Laser-cut acrylic sheets may have sharp edges that can cause cuts during handling or assembly	Deburring edges, careful handling	4	3	12
11	Eye strain from bright LEDs	Direct exposure to bright LEDs at close range may cause discomfort or temporary eye strain.	Do not look directly at LEDs during operation, limit exposure time	3	3	9
12	PCB Damage during handling	Physical stress or electrostatic discharge may damage components or traces, leading to the PCB failing	Careful handling, Electro-static discharge awareness	4	2	8
13	Overheating of LEDs	Excess current or poor design may generate heat, reducing component lifespan or causing failure	Correct resistor sizing, tesing	4	2	8
14	LED brightness inconsistency	Variegations between LEDs, any cause uneven brightness across digits, reducing display quality	Use matched components, testing	2	3	6
15	Loose electrical connections	Poor wiring or header connections may cause intermittent faults or unreliable LED control	Secure connections, srain relief	2	4	8